

Business Requirements Document

RFID Implementation

v0.01

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Version Control

Version	Date	Updated By	Comments
0.01	01/08/26	Glen Gregorio	Initial draft
0.02	01/19/26	Glen Gregorio	Added notes considering learnings from Zara store visit.
0.03	01/30/26	Glen Gregorio	Additional note with consideration to WHSE

Approval List

Version	Approver	Role	Approval Date
0.01	Cesar Catalan Jr.	Chief Information Officer	
0.01	Tristan De Jesus	Head of IT Projects and Delivery	
0.01	Gabriel Bustamante	Head of IT Systems Operations	
0.01	Chris Visaya	Head of IT Enterprise Application	
0.01	Ralph Mallari	Head of IT Infrastructure	

Executive Summary

Provide a high-level overview of the project, including the business opportunity, objectives, and expected benefits.

The company intends to implement **Radio Frequency Identification (RFID)** technology across its retail ecosystem to improve inventory accuracy, operational efficiency, and visibility across warehouses and stores. This initiative will integrate RFID hardware and middleware with the existing **SAP ERP** landscape, enabling automated, near-real-time inventory tracking and event-driven business processes.

The implementation will cover **inbound, outbound, internal movement, sales, and return** scenarios across warehouses, stores, and selected office locations. RFID events will be logically aggregated and translated into meaningful business events before being posted to SAP, ensuring scalability, performance, and data accuracy.

Expected benefits include:

- Improved **inventory accuracy and visibility** across warehouses and stores
- **Reduced labor effort and cycle count time**
- Faster inventory reconciliation and exception detection
- Enhanced customer experience via accurate product availability
- Stronger readiness/support for omnichannel fulfillment

Business Objective

List the business goals this project aims to fulfill.

1. Achieve inventory accuracy across warehouses and stores.
2. Reduce manual scanning and counting effort in stores and warehouses
3. Enable near-real-time visibility of inventory movement
4. Improve store shelf availability and replenishment responsiveness
5. Support faster and more reliable store receiving and sales return processing
6. Provide a scalable RFID foundation that supports future rollout to additional stores and warehouses

Background and Current State

Describe the current business process and the problem or opportunity prompting the project.

Current State

Currently, inventory movements are tracked primarily through barcode scanning and manual processes within SAP. Store cycle counts and stock validation are labor-intensive, time-consuming, and prone to human error. Inventory discrepancies are often discovered late, impacting replenishment, availability, and customer satisfaction.

Inbound and outbound processes rely heavily on manual confirmation steps. Store operations, including receiving, shelf replenishment, and returns, lack automated validation mechanisms. Visibility into shelf-level availability versus backroom stock is limited.

Opportunity

RFID technology provides an opportunity to automate inventory capture, reduce manual intervention, and enable continuous inventory awareness. Observations from mature RFID retail implementations (e.g., fashion retailers) indicate significant improvements in cycle count speed, store accuracy, and operational efficiency when RFID is embedded into daily store workflows.

Scope of Work

Identify what is within and out of scope of the solution coverage.

In Scope

- **RFID Events Management**
 - Capture RFID tag reads from fixed and handheld readers
 - Aggregate raw RFID reads into logical business events using defined time windows and filtering rules
 - Support logical grouping of readers into business zones (e.g., warehouse inbound dock, warehouse outbound dock, store backroom, sales floor, POS)
 - Generate standardized RFID business events (e.g., **Goods Receipt, Goods Issue, Stock Transfer, Sale, Return**)
- Logical Zoning & Reader Grouping
 - Support logical grouping of RFID readers into business zones such as:
 - **Warehouse inbound docks**
 - **Warehouse outbound docks**
 - **Staging / buffer zones**
 - **Store backroom receiving**
 - **Sales floor, POS checkout, and exit zones**
 - Apply business rules at the **zone level**, not per individual reader
 - Prevent cross-zone misreads (e.g., inbound reads interpreted as outbound)
- Warehouse Operations:
 - **Inbound**: Supplier deliveries (local & foreign), warehouse transfers, returns to vendor, returns from sales
 - **Outbound**: Shipments to stores, sales fulfillment, transfers to head office
- Store Operations:
 - **Store receiving** via backroom RFID zones
 - **Separation** of backroom inventory and sales-floor availability

- RFID-enabled cycle counting (full and partial)
 - Shelf replenishment support based on inventory visibility
 - Sales and checkout inventory decrement (RFID-assisted or validated)
 - Customer sales returns processing
- RFID hardware deployment (tags, fixed/mobile readers)
- SAP Integration
 - Integration of aggregated RFID business events into SAP (e.g., MM, SD, EWM, SAP Retail)
 - Posting of inventory movements using bulk and aggregated transactions
 - Support for error handling, retries, and reconciliation
 - Auditability of RFID-triggered postings
- Redesign affected processes:
 - Tagging of RFID Tags
 - Receiving and put-away
 - Picking, packing, shipping
 - Stock replenishment, cycle counting, returns
- Reporting and Monitoring
 - Operational dashboards for RFID events and exceptions
 - Visibility into inventory accuracy, cycle count performance, and exception rates.
- Exception Handling & Governance
 - Classification of RFID exceptions (missing reads, extra reads, unexpected exits, duplicate detections)
 - Role-based manual override capabilities
 - Full audit trail for overrides and adjustments
 - Clear ownership of exception resolution (store vs warehouse vs HQ)
- User training and change management

Out of Scope

- Core SAP ERP redesign
- Non-inventory use cases (marketing/customer tracking)
- RFID for non-retail business units or external partners
- Full replacement of barcodes where still required
- Advanced analytics beyond basic inventory updates

Key considerations:

RFID Event → SAP Action Matrix (Warehouse & Store)

Warehouse RFID Events → SAP Actions

RFID Business Event	Trigger Zone	Business Meaning	SAP Action	SAP Object	Posting Rule	Exception Handling
Inbound Receipt Detected	Warehouse Inbound Dock	Goods physically arrived at warehouse	Post Goods Receipt (GR)	Inbound Delivery / PO	Aggregated per shipment / ASN	Shortage, overage flagged for review
Unexpected Inbound Item	Warehouse Inbound Dock	Item not expected in shipment	No auto-posting	—	Logged as exception	Manual investigation required
Putaway Transition	Dock → Storage Zone	Item moved into storage	No SAP posting	—	Informational only	None
Inventory Count Completed	Storage Zone	RFID-assisted cycle count	Inventory Difference Proposal	Inventory Doc	No auto-post	Approval required
Outbound Ready Detected	Staging / Outbound Zone	Shipment prepared	No SAP posting	—	Informational	Missing items flagged
Outbound Exit Confirmed	Warehouse Outbound Dock	Goods left warehouse	Post Goods Issue (PGI)	Outbound Delivery	Aggregated per delivery	Partial shipment exception
Warehouse-to-Warehouse Transfer	Outbound / Inbound Dock	Stock transfer between warehouses	STO PGI / GR	Stock Transfer Order	Event-paired	Mismatch flagged
Return to Vendor Exit	Warehouse Outbound Dock	RTV shipment dispatched	Post Goods Issue (RTV)	Return PO	Aggregated	Validation required

Store RFID Events → SAP Actions

RFID Business Event	Trigger Zone	Business Meaning	SAP Action	SAP Object	Posting Rule	Exception Handling
Store Receiving Detected	Store Backroom	Goods arrived at store	Post Goods Receipt	STO / Delivery	Aggregated per delivery	Missing / extra flagged
Unexpected Store Receipt	Store Backroom	Item not on delivery	No auto-posting	—	Logged	Investigation required
Backroom to Sales Floor	Sales Floor Entry	Item available for sale	No SAP posting	—	Visibility only	None
Store Cycle Count Completed	Store Zones	RFID inventory count	Inventory Adjustment Proposal	Inventory Doc	No auto-post	Store approval
POS Sale Confirmed	POS / Exit Zone	Item sold	Reduce Inventory	Billing Doc	POS-driven	RFID Tag failure fallback
Unexpected Exit Detected	Store Exit	Item exited without sale	No auto-posting	—	Exception only	Loss prevention review
Customer Return Detected	Store Backroom / POS	Item returned	Post Goods Receipt	Return Doc	Validated	Quarantine if mismatch
Store-to-Warehouse Return	Store Outbound Zone	Stock returned to WH	PGI for Return STO	STO	Aggregated	Damage check

RFID Reader type needs:

- Handheld scanners: for cycle counts, stock checks, picking stocks and possibly exception handling
- Fixed scanner: docks, conveyors, packing stations, gates or checkout gates

RFID Tag informations:

- Unique item identifier (EPC - Electronic product code mapped to SAP article code or GTIN)

Assumptions and Constrains:

- RFID tags will carry a unique identifier linked to master data in SAP
- Product master, pricing, and business rules will remain in SAP
- WHSE | Store layouts and physical constraints may impact reader placement
- Staff training and change management are critical success factors

Reference Document

The following documents were used as references of this document.

Document Title	Document Type / Version
 RFID_Demo_Checklist_and_BRD_Enhancement	Google Sheet